

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup

Castalia

Maxx Seminario

University of Nebraska–Lincoln

September 22, 2026

Cancer Risk Overview

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

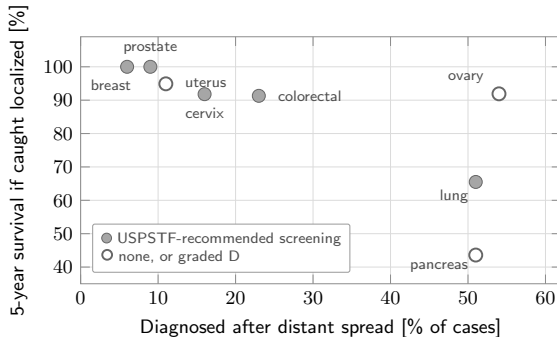
Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup



[1] SEER 21, 2016–2022; [2, 3] USPSTF

Why Ovarian: Late, Lethal, and Unscreened

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup



Site	Distant at dx	Deaths/100 cases	Screening (USPSTF)
Ovary	54 %	59	D
Pancreas	51 %	78	D
Lung	51 %	54	B, LDCT
Colorectal	23 %	35	A
Cervix	16 %	31	A
Uterus	11 %	21	none
Prostate	9 %	11	C, PSA
Breast	6 %	13	B

USPSTF: A/B recommended; C offered selectively; D recommended *against*; none: no statement [7].

[1] SEER, 2026 estimates; [2, 3] USPSTF

Most lethal gynecologic cancer [1]: 92 % survivable if caught localized, yet 54 % present after spread, and screening is graded D [2]. UKCTOCS moved stage, not mortality [4].

Concept Diagram

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup



Symptomatic-patient triage on a disposable four-site cartridge; four markers read at the same instant, fused on chip.

Four-Marker Test

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

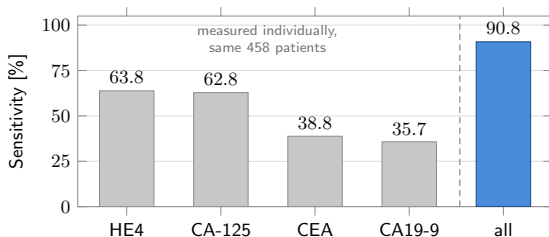
Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup



Sensitivities from 458 patients, each marker measured alone and then the four combined [5].

Marker	Molecule	Cut-off
CA-125	MUC16 mucin epitope [8]	35 U/mL
HE4	WFDC2 glycoprotein [6]	70 pmol/L
CA19-9	sialyl-Lewis ^a antigen	37 U/mL
CEA	CEACAM5 glycoprotein	5 ng/mL

SoC Block Diagram

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup



Physical Implementation

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

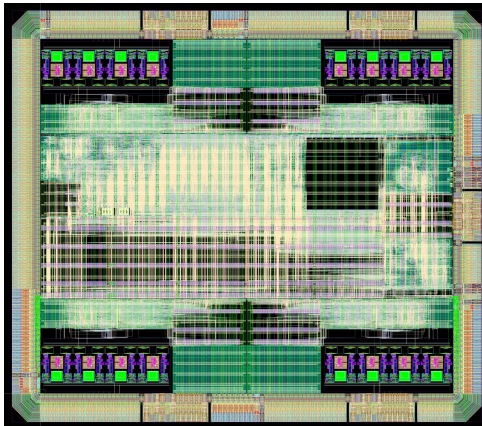
Digital
Subsystem

On-Chip
Inference

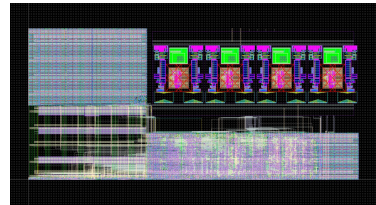
Tapeout

References

Backup



chip top, 2.25×2.0 mm: two tile rows with their analog channels, the control plane between them, pad ring



one channel tile: TCM, analog channel, hart logic

Potentiostat Schematic

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

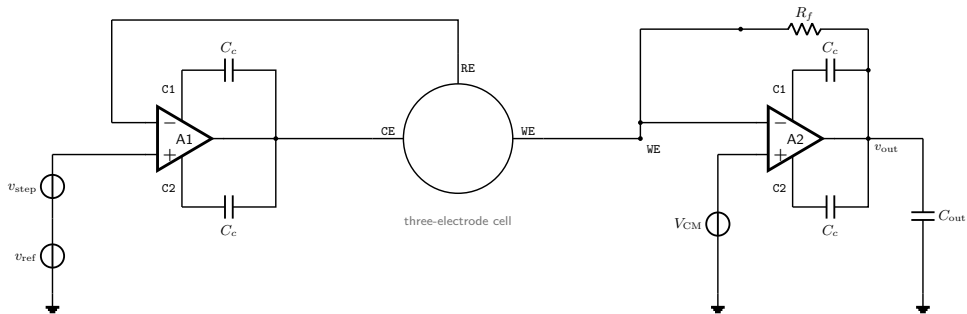
Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup



Analog Top-Level Schematic

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

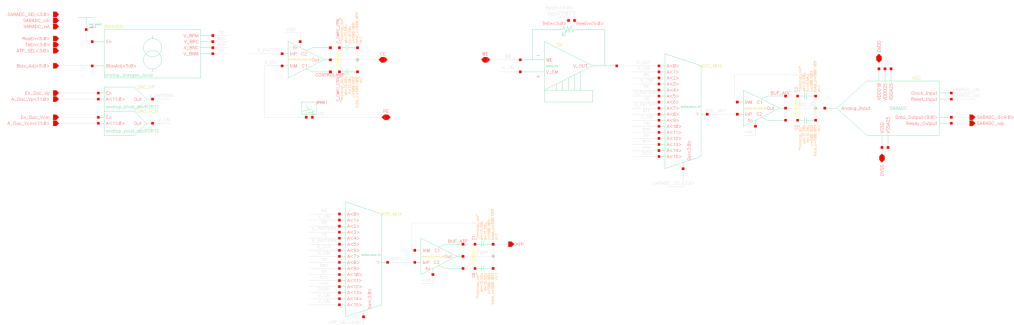
Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup



one channel: local bias generator, reference and v_{cm} DACs, A1, cell, A2 with R_f , 16:1 input multiplexer and buffer, SAR, analog test port

Electrode Model in Verilog-A

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

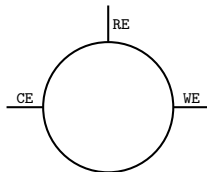
Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup



R_u, R_{ce}, C_{dl} lumped; the faradaic branch Z_f is the Verilog-A element.

i_f faradaic current
 η overpotential, $E - E^{0'}$
 f F/RT
 $C_{O,R}(0, t)$ surface conc. of O, R
 C^* bulk conc.
 δ Nernst layer

Kinetics $i_f = nFAk^0 [C_R(0, t) e^{(1-\alpha)n f \eta} - C_O(0, t) e^{-\alpha n f \eta}]$
 Transport $\partial C / \partial t = D \partial^2 C / \partial x^2$, finite-volume, 28 expanding slabs per species
 Boundaries flux = $\pm i_f / nFA$ at the surface; $C(\delta, t) = C^*$ at the Nernst layer

Parameter	Symbol	Default	Unit	Quantity
E0	$E^{0'}$	0.0	V	formal potential of the couple
nel	n	1	—	electrons transferred
alpha	α	0.5	—	transfer coefficient
k0	k^0	0.1	cm s^{-1}	standard heterogeneous rate constant
Dif	D	5×10^{-6}	$\text{cm}^2 \text{s}^{-1}$	diffusion coefficient, both species
Area	A	0.01	cm^2	electrode area
Cox	C_O^*	1×10^{-6}	mol cm^{-3}	bulk concentration, oxidised species
Cred	C_R^*	0	mol cm^{-3}	bulk concentration, reduced species
Temp	T	298.15	K	interface temperature
expmax	—	40	—	exponent clamp on the kinetic terms
gpar	—	1×10^{-15}	S	DC shunt across the terminals

Equations: Bard–Faulkner [13]. Discretisation: Britz–Strutwolf, Feldberg [17, 18]. Precedent for Verilog-A electrode co-simulation: Jog et al. [19].

Emulating the Four Markers

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup

Marker	Reporter	$E^{0'}$ (V)	Peak (nA)	C_R^* (μ M)	Cut-off
CA-125	methylene blue	-0.15	856	279	35 U/mL [5, 8]
HE4	ferrocene	+0.10	1311	394	70 pmol/L [6]
CA19-9	TMB / HRP [20]	+0.25	1659	458	37 U/mL [5]
CEA	anthraquinone	+0.40	939	303	5 ng/mL [5, 14]

$E^{0'}$ reporter formal potential, i.e. peak position; chosen in the -0.6 to $+0.4$ V window [22, 23], 150–250 mV apart, not measured per reporter

Peak inside 50 nA to 5 μ A on screen-printed electrodes [14, 15, 21]; C_R^* fitted to reach it

k^0 0.1 cm/s, reversible; nothing published on carbon for these markers

Shared $\alpha = 0.5$, $n = 1$, $D = 5 \times 10^{-6}$ cm²/s, $A = 0.01$ cm², 37 °C, $R_f = 560$ k Ω

Representative values, not marker-measured: these slides demonstrate the channel, not an assay.

Potentiostat Range

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

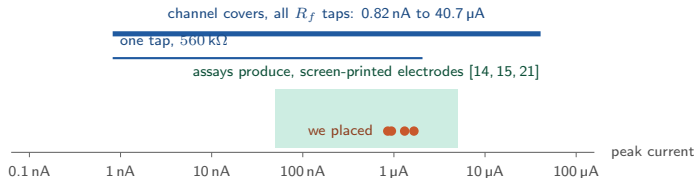
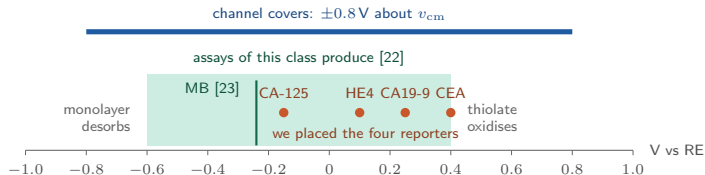
Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup



Potentials: spaced 150–250 mV for one-sweep resolution inside the reporter window, not per-reporter values. Peak heights: inside the published envelope and one tap's full scale, then C_R^* fitted. Silicon is unaffected: the channel covers the assay band with 0.2–0.4 V and a decade of current to spare, so moving the dots changes what the simulation shows, not what the chip can measure.

Differential-Pulse Voltammetry

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

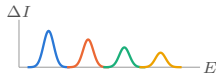
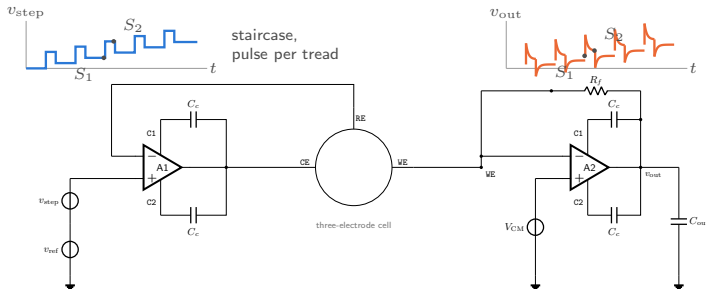
Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup



firmware, per tread:
 $\Delta I = I(S_2) - I(S_1)$ against E
 one peak per marker

Excite
Sense
Sample
Resolve

DAC: staircase with a pulse on every tread, applied at RE by A1

$$v_{out} = v_{cm} - i_{WE} R_f$$

SAR at S_1 (before the pulse) and S_2 (end of the pulse), once charging has decayed

firmware: $\Delta I = I(S_2) - I(S_1)$ vs E ; peak position: marker, peak height: concentration

DPV Simulation Results

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

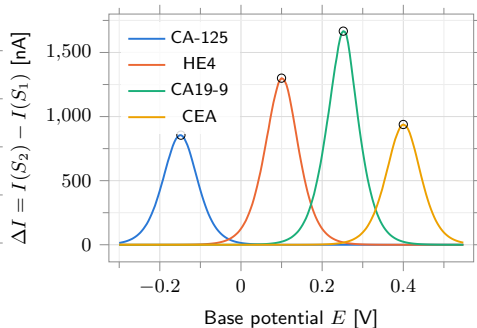
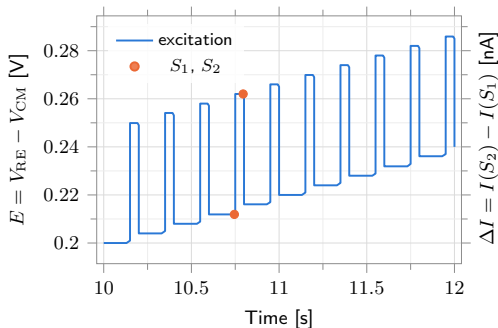
Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup



Peak position identifies the reporter, peak height carries concentration. Apexes within 2 mV and 1 % of the assay targets; blank below 1 pA.

Dose-Response Simulation Results

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

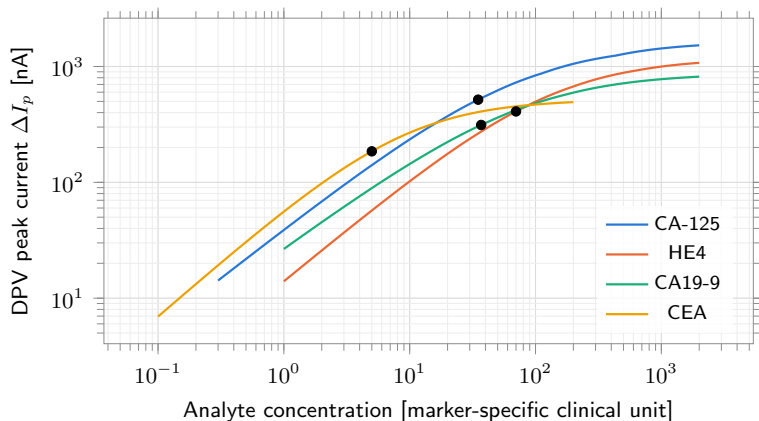
Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup



CA-125 35 U/mL, HE4 70 pmol/L, CA19-9 37 U/mL, CEA 5 ng/mL: all four on the steep part of the curve at one R_f setting.

Cyclic Voltammetry, Spectre Simulation

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

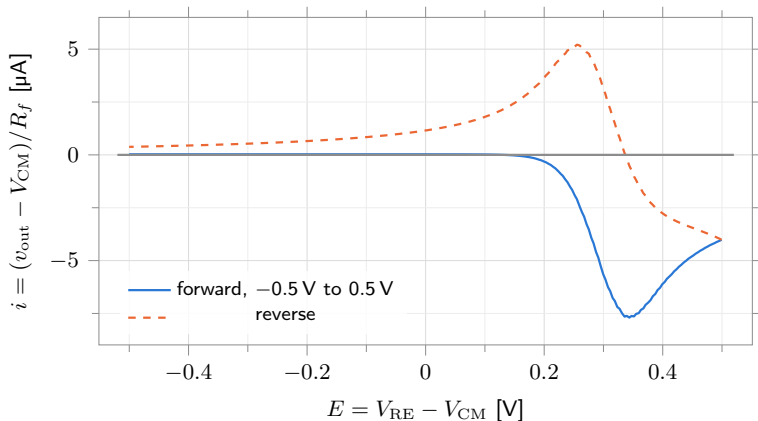
Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup



Staircase potential from the on-chip DAC, current digitised by the on-chip SAR, unclipped over the sweep.

Electrochemical Impedance Spectroscopy Motivation

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup

cartridge



single-shot sandwich immunoassay, incubate and read once

wearable

continuous glucose monitors, 10–15 days [27]
wound patch [28]; aptamer sensor, one week in vivo [29]



implant

Eversense implanted glucose monitor, 180–365 days [27]



1 min

1 h

1 day

1 week

1 month

1 year

time the electrode spends in the sample

EIS gives
Firmware

each site's $R_{ct} \parallel C_{dl}$ arc, trended over the electrode's life, from the same channel
re-ranges, recalibrates, or retires the site

Drift is systematic: averaging removes noise, not bias [25].

Exercising EIS

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

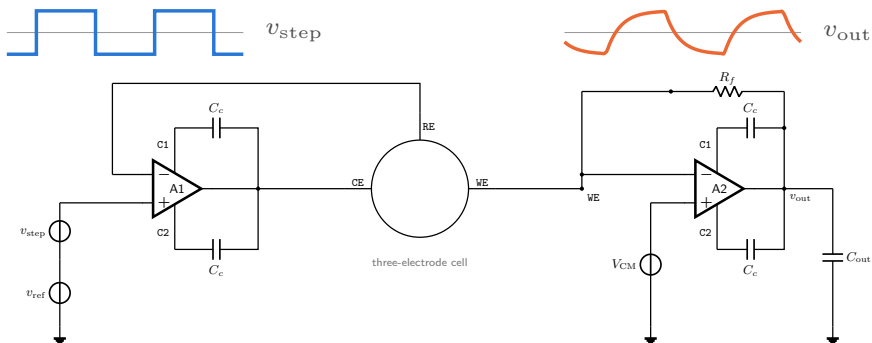
Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup



Excite
Sense
Digitise
Solve

DAC: a small two-code square about the DC setpoint v_{ref}
 v_P across the cell; $i_{WE} = -v_{\text{out}}/R_f$
 the SAR records both; firmware I/Q demodulation
 $\hat{Z} = -R_f \hat{V}_P / \hat{V}_{\text{OUT}}$, or a ratio to Z_{cal}

EIS I/Q Demodulation

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

Digital
Subsystem

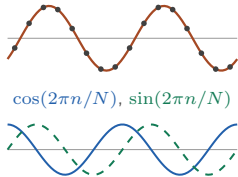
On-Chip
Inference

Tapeout

References

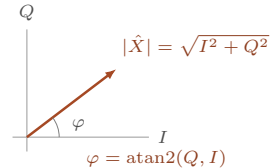
Backup

record c_n , N per period, M periods



$$I = \frac{2}{MN} \sum c_n \cos \frac{2\pi n}{N}$$
$$Q = \frac{2}{MN} \sum c_n \sin \frac{2\pi n}{N}$$

Records v_P and v_{out}
Divide $\hat{Z} = -R_f \hat{V}_P / \hat{V}_{OUT}$
Ratio against Z_{cal} : R_f drops out



EIS Test Loads

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

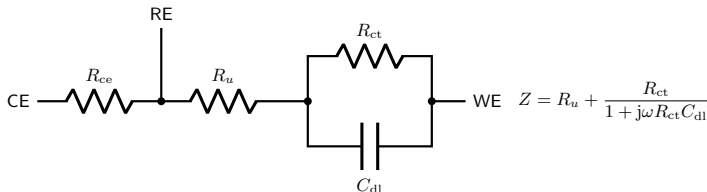
Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup



Load	R_{ct}	C_{dl}	Gain code	Stands for
resistor	–	–	$N = 0$	an 11 k Ω resistor in place of the cell: no interface, so the trace is the channel's own error
fouled	10 k Ω	10 μ F	$N = 0$	a degraded electrode: adsorbed protein and biofilm open leakage paths and enlarge the interface
typical	100 k Ω	1 μ F	$N = 20$	an electrode in use
fresh	1 M Ω	100 nF	$N = 63$	a new, clean surface

$R_u = R_{ce} = 1 \text{ k}\Omega$ throughout. Fouling: R_{ct} falls and C_{dl} rises through an electrode's life; tracking that drift is what EIS is for here.

EIS Simulation Results

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

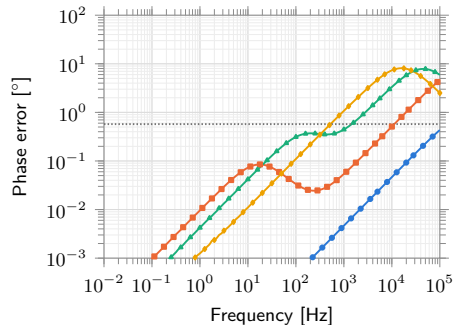
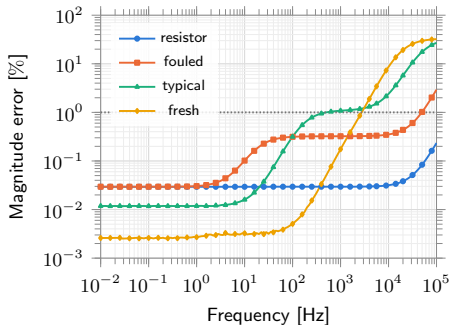
Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup



Accuracy $\pm 1\%$, $\pm 0.6^\circ$ to 5 kHz; $\pm 5\%$ to 27 kHz. Limit: A1's loop bandwidth
 Range 246Ω to $12.1\text{ M}\Omega$. Limits: TIA compliance below, one LSB of output above; 320 R_f codes in between

EIS Errors

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

Digital
Subsystem

On-Chip
Inference

Tapeout

References

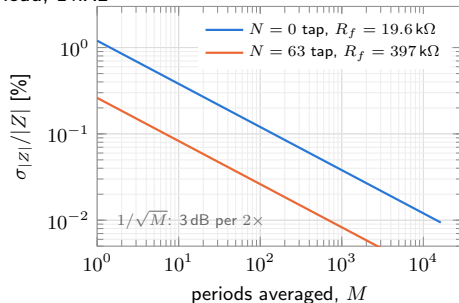
Backup

$$\varepsilon_{|Z|} \approx \underbrace{\frac{R_f/|Z|}{A_{\text{eff}}}}_{\text{systematic}} + \underbrace{\frac{i_n |Z|}{V_1} \sqrt{\frac{2}{NM}}}_{\text{random}}$$

$R_f/ Z $	feedback resistor over the load: the gain code chosen
A_{eff}	effective loop gain, ≈ 76 dB measured (0.66 % at $R_f/R = 40$); $A_{\text{OL}} \approx 83$ dB
i_n	front-end current noise per sample: 273 nA at $N=0$, 61 nA at $N=63$
V_1	excitation fundamental, $\frac{4}{\pi} \frac{A}{2}$
N, M	samples per period, periods averaged

The systematic term is fixed for a gain code and load, and the Z_{cal} ratio removes it. The random term is what remains, and it falls as $1/\sqrt{M}$.

Random term against periods averaged, 11 k Ω load, 1 kHz



EIS Calibration

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup

$$\hat{Z} = Z_{\text{cal}} \frac{(\hat{V}_P / \hat{V}_{\text{OUT}})}{(\hat{V}_P / \hat{V}_{\text{OUT}})_{\text{cal}}} \quad \text{same gain code, same frequency}$$

	On-chip poly, per decade	Off-chip standard, at test
Cancels	R_f , its pole, DAC, LSB, loop-gain floor	the same
Known to	$\sim 20\%$ chip to chip	$< 0.1\%$
Run	any time, in the field	once, stored in flash
Good for	trends: fouling is a change in one electrode's Z	absolute ohms

General-Purpose Class-AB Amplifier

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

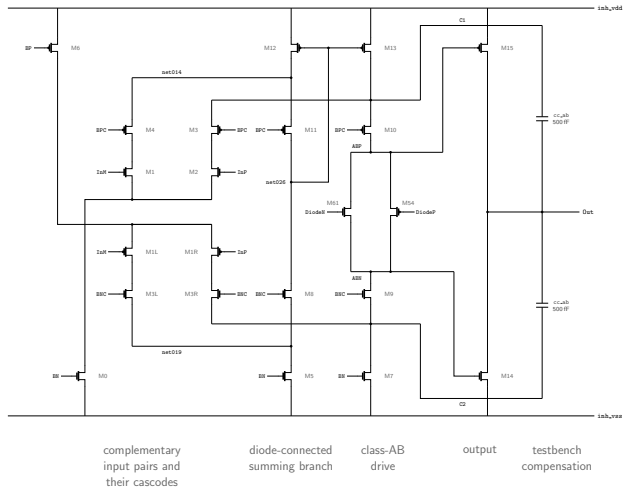
Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup



Class-AB Amplifier Results

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup

Parameter	Unit	tt	ff	fs	sf	ss	σ (MC)
Open-loop gain	dB	82.77	79.88	83.14	83.37	84.94	1.06
Phase margin	deg	73.52	74.70	74.04	72.86	73.34	1.42
Dominant pole	kHz	1.592	2.286	1.538	1.494	1.225	0.276
Gain-bandwidth product	MHz	22.78	23.19	22.89	22.94	22.59	3.57
Input offset voltage	mV	0.622	0.649	0.624	0.569	0.456	5.201

Bipolar Potentiostat Results

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

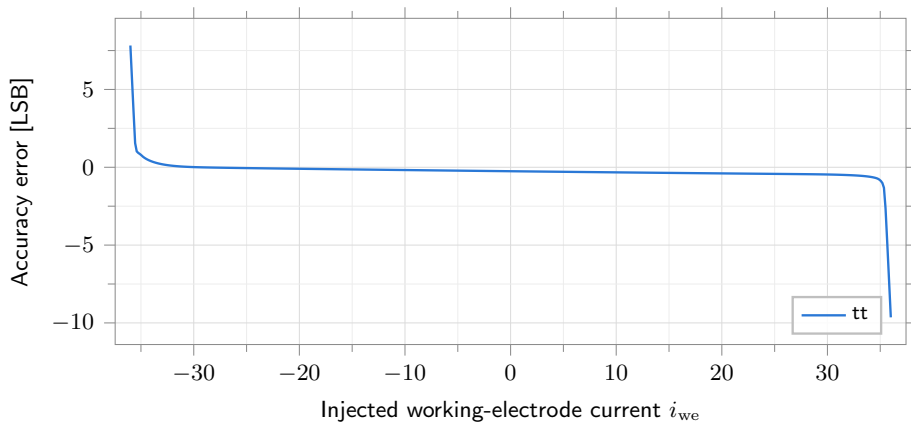
Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup



Programmable Feedback Resistor Results

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

Digital
Subsystem

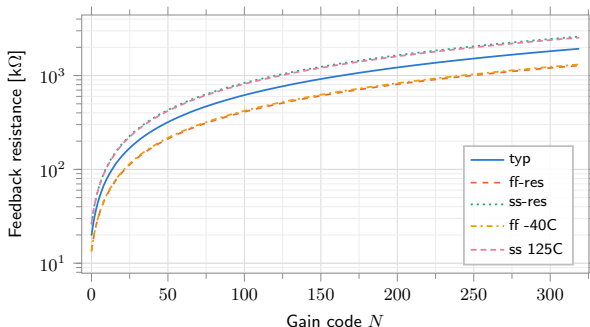
On-Chip
Inference

Tapeout

References

Backup

- $R_f = 18 \text{ k}\Omega + 6 \text{ k}\Omega \cdot N$; full scale = $0.80 \text{ V}/R_f$, LSB = $1.59 \text{ mV}/R_f$
- **Max $\pm 40.7 \mu\text{A}$, min full scale $\pm 0.41 \mu\text{A}$, resolution 0.82 nA**
- Noise floor $i_n^2 = e_n^2 (1 + R_f/|Z_{el}|)^2/R_f^2 + 4kT/R_f$: 341 pA alone, 3.6–21 nA into an electrode (0.1–100 Hz, 560 k Ω)



N	R_f	Full scale	LSB
0	19.6 k Ω	$\pm 40.7 \mu\text{A}$	81 nA
63	397 k Ω	$\pm 2.02 \mu\text{A}$	4.0 nA
319	1.93 M Ω	$\pm 0.41 \mu\text{A}$	0.82 nA

Mismatch σ	σ/typ	
$R(N=0)$	71 Ω	0.36 %
$R(N=319)$	0.7 k Ω	0.04 %
Seam 63→64	504 Ω	6.8 %

10-Bit SAR Transfer Function

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

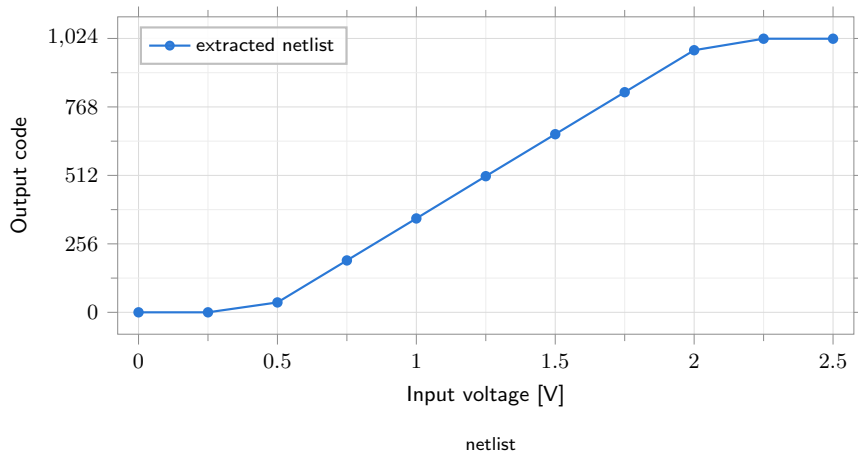
Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup



■ extracted

SAR ADC Results

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup

Parameter	Unit	tt	ff	fs	sf	ss
Fitted slope	codes/V	628.59	628.59	628.40	628.73	628.13
Effective LSB	mV	1.5909	1.5909	1.5914	1.5905	1.5920
Input at code 0	V	0.4408	0.4408	0.4407	0.4408	0.4399
Usable span	V	1.6274	1.6274	1.6280	1.6271	1.6286
Worst residual	LSB	0.495	0.495	0.490	0.477	0.484
RMS residual	LSB	0.294	0.294	0.277	0.288	0.302
Analog supply current	μ A	6.36	4.79	5.45	8.44	15.43
Energy per conversion	pJ	20.8	19.3	18.1	27.3	49.5

DAC Results

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup

Parameter	Unit	tt	ff	fs	sf	ss	σ (MC)
Full-scale output	V	2.499388	2.499388	2.499388	2.499388	2.499388	0.000001
LSB size	μV	610.351	610.351	610.351	610.351	610.351	0.0003
INL, most positive	LSB	0.113	0.101	0.100	0.127	0.130	0.610
INL, most negative	LSB	-0.376	-0.338	-0.390	-0.362	-0.414	0.620
DNL, most positive	LSB	0.002	0.001	0.002	0.002	0.002	1.128
DNL, most negative	LSB	-0.423	-0.374	-0.424	-0.424	-0.479	1.415

Comparison Against the State of the Art

Castalia

Maxx Seminario

	Castalia (this work)	Myshkin [9]	Chen '19 [10]	Ghoreishizadeh '17 [11]	Lin '23 [12]
Target Application	Modality	CA/CV	CA/CV	CA/CV	CV/EIS
SoC Overview	Channels	1	1	1	1
Analog Front End	Current direction	unipolar	bipolar	bipolar	bipolar
Measurement Modes	$I_{\max}$ (μA)	+20	± 5	± 20	-48/+16
Analog Block Results	Resolution (pA rms)	56.5	41	0.47	1.7
Digital Subsystem	ADC step (pA)	153	n/a	n/a	n/a
On-Chip Inference	Converter	12-bit dual-slope	TDC	ADC	-
Tapeout	Power (mW)	3.0	0.016	9.3	1.0
References	Process (nm)	65	180	350	180
Backup	On-chip processor	1 RISC-V hart	none	none	none
	On-chip ML	NPU, 1 mode	none	none	none

^aInput-referred noise, 0.1–100 Hz, ideal R_f , no electrode.

^bOne converter code at the maximum-gain tap; 81 nA at the lowest.

^cOne channel measuring: orchestrator + one tile (7.6 mW digital) + one analog channel (1.3 mW); the other tiles and channels gated, pads quiet.

Analog Power Distribution

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

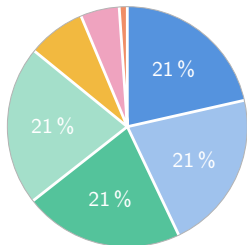
Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup



- A1, control amplifier, 244 μ W, 21 %
- A2, transimpedance amplifier, 244 μ W, 21 %
- ADC input buffer, 244 μ W, 21 %
- debug buffer, 244 μ W, 21 %
- two R-2R ladders, 89 μ W, 8 %
- local bias generator, 59 μ W, 5 %
- SAR converter, 12 μ W, 1 %

Power

One channel	1.32 mW
Four channels	5.3 mW

SoC Block Diagram

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

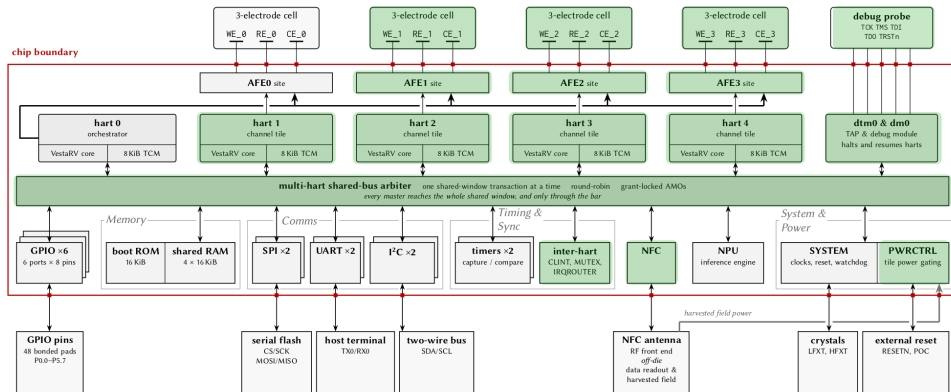
Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup



■ new since Myshkin

SoC Specifications

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup

Digital

Harts	5 (1 orchestrator, 4 channel tiles)
ISA	rv32imac_zba_zbb_zbs_zbc
Clock	24 MHz rated, closed at 25 MHz
Measured CPI	1.260 aggregate, nine benchmarks
Private TCM	8 KiB per hart, 1-cycle
Shared RAM	80 KiB behind a round-robin arbiter
Synchronisation	16 hardware mutexes, LR/SC, AMOs
Interrupts	claim/complete, any vector to any hart
Accelerator	NPU: MLP, conv-1D, XNOR, GEMM
Power domains	4 MTCMOS, one per channel tile
Debug	JTAG, RISC-V Debug Module
Field boot	NFC ISO 14443A harvested boot

Analog and physical

Measurement sites	4 identical, simultaneous
Topology	bipolar grounded-WE potentiostat
Electrode pads	12 (CE/RE/WE $\times$ 4)
Converter	10-bit SAR per channel, extracted
Reference DAC	12-bit R-2R per channel
Feedback resistor	$18\text{ k}\Omega + 6\text{ k}\Omega \cdot N$, 320 codes
Full scale	$\pm 40.7\text{ }\mu\text{A}$ to $\pm 0.41\text{ }\mu\text{A}$
EIS	firmware mode of the same channels
Analog supply	2.5 V , $v_{\text{cm}} = 1.25\text{ V}$
Process	65 nm CMOS, 1P8M, HVT cells
Die	$2.25 \times 2.0\text{ mm}$, MPW
Digital power	12.8 mW core post-route, SS, 25 MHz

Digital Results

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup

Digital, post-route

Internal	8.69 mW
Switching	9.99 mW
Leakage	4.70 mW

Core logic	12.8 mW
Pad ring*	10.5 mW
Total	23.4 mW

One channel tile

Total	1.742 mW
Gated	0

Measured CPI, timed kernels

Benchmark	Instr.	CPI
spmv	802 447	1.143
multiply	20 892	1.163
vvadd	2 408	1.374
median	4 015	1.398
qsort	123 205	1.398
rsort	146 548	1.412
dhrystone	200 523	1.504
towers	3 747	1.682
memcpy	11 017	1.726

Aggregate	1 314 802	1.260
------------------	------------------	--------------

SS 0.9 V 125 °C, 25 MHz, statistical activity 0.2. *Pads at the statistical activity.

Digital Power Gating

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

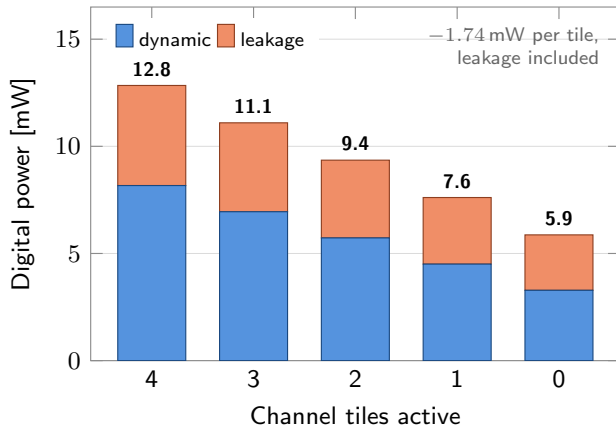
Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup



SoC Power Modes

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup

Mode	Digital	Analog	Total	What is off
Everything on, pads toggling	23.4 mW	5.3 mW	28.7 mW	nothing; pads at the statistical activity
Everything on, pads quiet	12.8 mW	5.3 mW	18.1 mW	pad switching; pad leakage is 0.03 mW
One channel measuring	7.6 mW	1.3 mW	8.9 mW	three tiles gated, with their channels
Orchestrator only	5.9 mW	0	5.9 mW	all four tiles and channels gated
Orchestrator asleep	2.6 mW	0	2.6 mW	its clock; always-on leakage remains
Deep sleep	< 2.6 mW	0	< 2.6 mW	shared RAM banks and NPU RAM powered off; NPU RAM alone leaks 0.66 mW

Digital: SS 0.9 V 125 °C, 25 MHz, statistical activity 0.2.

SoC Power Distribution

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

Digital
Subsystem

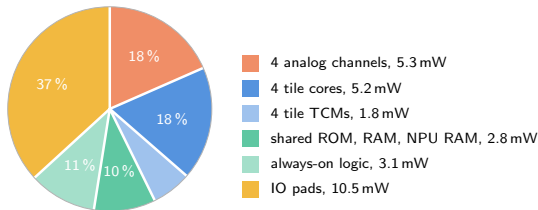
On-Chip
Inference

Tapeout

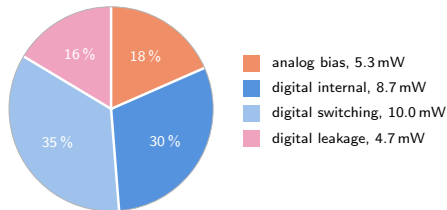
References

Backup

By block, 28.6 mW



By type, 28.7 mW



Everything on. Digital: SS 0.9 V 125 °C, 25 MHz, statistical activity 0.2, pads included. Analog: 1.32 mW per channel, nominal.

Low-Power Inference, Cancer Risk

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup

$$s = \sigma \left(\sum_{k=1}^4 w_k \hat{c}_k + b \right)$$

Input	peak current, peak potential and baseline from each of the four sites
Mode	dense layer, one sigmoid; the logistic form of ROMA [6], four inputs
Weights	trained off-chip, staged by firmware
Output	one risk score

Tapeout

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup



Services About Us Schedule IP University FinFET Program FAQ 日本語

TSMC MPW SHARED TAPEOUT SCHEDULE

Type to search	Search					
Tech (nm)	▲ Flavor	▲ Trial	Final	Tapeout	▲ Est. Ship	
28	HPC+ RF	08/26/2026	09/02/2026	09/09/2026	12/17/2026	
180	MS RF G	09/02/2026	09/09/2026	09/16/2026	11/09/2026	
65	MS RF LP	09/02/2026	09/09/2026	09/16/2026	12/13/2026	
16	FFC TUIFP	09/22/2026	09/29/2026	10/06/2026	01/17/2027	
180	MS RF G	09/30/2026	10/07/2026	10/14/2026	12/16/2026	
65	MS RF GP	10/07/2026	10/14/2026	10/21/2026	01/17/2027	
180	HV BCD Gen2	10/14/2026	10/21/2026	10/28/2026	01/21/2027	
180	MS RF G	10/28/2026	11/04/2026	11/11/2026	01/17/2027	
28	HPC+ RF	10/28/2026	11/04/2026	11/11/2026	02/18/2027	
65	MS RF LP	11/04/2026	11/11/2026	11/18/2026	02/14/2027	
180	MS RF G	11/18/2026	11/25/2026	12/02/2026	02/03/2027	
28	HPC+ RF	11/25/2026	12/02/2026	12/09/2026	03/18/2027	
180	HV BCD Gen2	12/02/2026	12/09/2026	12/16/2026	03/11/2027	
65	MS RF GP	12/02/2026	12/09/2026	12/16/2026	03/14/2027	

Foundry, process	TSMC 65 nm GP, MUSE shared MPW
Selected run	65 MS RF GP: trial GDS 2 Dec, final 9 Dec , tapeout 16 Dec 2026, ship 14 Mar 2027
Die	2.25 × 2.0 mm; Y fixed at 2 mm, priced per area
Silicon	\$25 520
Budget	\$30 000 with tax and packaging

References (1/3)

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup

- [1] NCI, *SEER Cancer Stat Facts*, per-site pages, seer.cancer.gov/statfacts, accessed 20 Sep. 2026. Stage, survival: SEER 21, 2016–22; cases, deaths: 2026 est.
- [2] D. C. Grossman *et al.*, "Screening for ovarian cancer: USPSTF recommendation statement," *JAMA* 319:588, 2018.
- [3] USPSTF, *A and B Recommendations; Prostate Cancer: Screening*, 2018; *Pancreatic Cancer: Screening*, 2019. uspreventiveservicestaskforce.org.
- [4] U. Menon *et al.*, "Ovarian cancer population screening and mortality after long-term follow-up in UKCTOCS," *Lancet* 397:2182, 2021.
- [5] F. Chen *et al.*, "Clinical analysis of four serum tumor markers in 458 patients with ovarian tumors: combined use of HE4, CA125, CA19-9, and CEA," *Cancer Manag. Res.* 10:1313, 2018.
- [6] R. G. Moore *et al.*, "A novel multiple marker bioassay utilizing HE4 and CA125 for the prediction of ovarian cancer in patients with a pelvic mass," *Gynecol. Oncol.* 112:40, 2009.
- [7] USPSTF, *Grade Definitions* (current, after July 2012), uspreventiveservicestaskforce.org, accessed 20 Sep. 2026.
- [8] P. Charkhchi *et al.*, "CA125 and ovarian cancer: a comprehensive review," *Cancers* 12:3730, 2020.
- [9] M. A. Seminario, S. McRobert, A. Uerling, P. Aberson, S. Balkir, J. A. Schmitz and E. J. Markvicka, "An electrochemical sensing SoC for autonomous wound monitoring," *IEEE ISCAS*, 2026.
- [10] Y.-C. Chen, S.-Y. Lu and Y.-T. Liao, "A microwatt dual-mode electrochemical sensing current readout with current-reducer ramp waveform generation," *IEEE TBioCAS* 13(6):1163, 2019.

References (2/3)

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup

- [11] S. S. Ghoreishizadeh *et al.*, "A differential electrochemical readout ASIC with heterogeneous integration of bio-nano sensors for amperometric sensing," *IEEE TBioCAS* 11(5):1148, 2017.
- [12] Y.-J. Lin *et al.*, "A multimodality electrochemical and impedance spectroscopy system-on-a-chip with temperature sensing and impedance-boosting techniques," *IEEE TBioCAS* 17(4):857, 2023.
- [13] A. J. Bard and L. R. Faulkner, *Electrochemical Methods: Fundamentals and Applications*, 2nd ed., Wiley, 2001 (Eqs. 3.3.11, 4.4.2, 4.4.29, Sec. 1.4.2, Eq. 6.2.19).
- [14] M. S. Wilson and W. Nie, "Multiplex measurement of seven tumor markers using an electrochemical protein chip," *Anal. Chem.*, 2006.
- [15] A. Kovářová *et al.*, "Biosensor with electrochemically active nanocomposites for signal amplification and simultaneous detection of three ovarian cancer biomarkers," *Electrochim. Acta*, 2023.
- [16] V. Mirčeski *et al.*, "Square-wave voltammetry: a review on the recent progress," *Electroanalysis*, 2013.
- [17] D. Britz and J. Strutwolf, *Digital Simulation in Electrochemistry*, 4th ed., Springer, 2016 (Ch. 3 box method, Ch. 7 expanding grid).
- [18] S. W. Feldberg, "Digital simulation: a general method for solving electrochemical diffusion-kinetic problems," *Electroanal. Chem.* 3:199, 1969; J. E. B. Randles, *Discuss. Faraday Soc.* 1:11, 1947.
- [19] Jog *et al.*, Verilog-A three-electrode model with kinetics and mass transport for co-simulation with interface circuits, *IEEE SENSORS*, 2024, doi:10.1109/SENSORS60989.2024.10784491; S. Madec *et al.*, *IEEE TCAS-I* 66(6):2188, 2019.
- [20] P. Fanjul-Bolado, M. B. González-García and A. Costa-García, "Amperometric detection in TMB/HRP-based assays," *Anal. Bioanal. Chem.* 382:297, 2005.

References (3/3)

Castalia

Maxx Seminario

Target

Application

SoC Overview

Analog Front End

Measurement

Modes

Analog Block

Results

Digital

Subsystem

On-Chip

Inference

Tapeout

References

Backup

- [21] M. Bilgi Kamaç *et al.*, "Point-of-care testing: a disposable label-free electrochemical CA125 and HE4 immunosensors for early detection of ovarian cancer," *Biomed. Microdevices* 25:18, 2023.
- [22] D. Kang, F. Ricci, R. J. White and K. W. Plaxco, "Survey of redox-active moieties for application in multiplexed electrochemical biosensors," *Anal. Chem.* 88(21):10452, 2016.
- [23] V. Clark, M. Aller Pellitero and N. Arroyo-Currás, "Explaining the decay of nucleic acid-based sensors under continuous voltammetric interrogation," *Anal. Chem.* 95(11):4974, 2023.
- [24] K. K. Leung *et al.*, "Elucidating the mechanisms underlying the signal drift of electrochemical aptamer-based sensors in whole blood," *ACS Sens.* 6:3340, 2021.
- [25] H. Li *et al.*, "Dual-reporter drift correction to enhance the performance of electrochemical aptamer-based sensors in whole blood," *J. Am. Chem. Soc.* 138:15809, 2016.
- [26] J. Sabatè del Río *et al.*, "An antifouling coating that enables affinity-based electrochemical biosensing in complex biological fluids," *Nat. Nanotechnol.* 14:1143, 2019.
- [27] Manufacturer labelling and FDA clearances: Dexcom G7 (10-day; 15-day, adults), Abbott FreeStyle Libre 3 (14 days), Senseonics Eversense 365 (implanted, 365 days; FDA clearance 17 Sep. 2024).
- [28] E. Shirzaei Sani *et al.*, "A stretchable wireless wearable bioelectronic system for multiplexed monitoring and combination treatment of infected chronic wounds," *Sci. Adv.* 9(12):eadf7388, 2023.
- [29] Son *et al.*, "Continuous, week-long, seconds-resolved in vivo drug measurements performed with a xenonucleic acid-employing electrochemical, aptamer-based sensor," *J. Am. Chem. Soc.* 148(9):10016, 2026.

Backup: Electrode Model Validation

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup

Scan rate mV s^{-1}	i_p Randles–Ševčík μA	i_p model μA	Error %	Error, $C_{\text{dl}} \rightarrow 0$ %	ΔE_p mV
25	0.951	0.952	+0.08	−0.45	58.78
50	1.345	1.349	+0.32	−0.42	59.85
100	1.902	1.915	+0.67	−0.38	60.41
200	2.690	2.721	+1.16	−0.33	62.33
400	3.804	3.875	+1.86	−0.25	64.74

Peak current against Randles–Ševčík and ΔE_p against Nicholson [13], reversible couple, $C_{\text{dl}} = 0$ column isolates the faradaic model.

Backup: USPSTF Screening Grades

Castalia

Maxx Seminario

Target
Application

SoC Overview

Analog Front End

Measurement
Modes

Analog Block
Results

Digital
Subsystem

On-Chip
Inference

Tapeout

References

Backup

Grade	Certainty of net benefit	Definition [7]	Cancer screening [2, 3]
A	high, substantial	Recommends the service. Offer or provide it.	cervix (2018), colorectal 50–75 (2021)
B	high moderate, or moderate	Recommends the service. Offer or provide it.	breast 40–74 (2024), lung LDCT 50–80 (2021), colorectal 45–49 (2021), prostate PSA 55–69 (2018)
C	at least moderate, small	Offer selectively, by professional judgment and patient preference.	
D	no net benefit, or harms outweigh	Recommends <i>against</i> the service. Discourage its use.	ovary (2018) , pancreas (2019)
I	cannot be determined	Evidence insufficient, poor, or conflicting.	—
none	—	No USPSTF statement exists for the condition.	uterus

Ovarian D is a verdict on a test that was tried, CA-125 plus transvaginal ultrasound [2, 4], not on the absence of one.